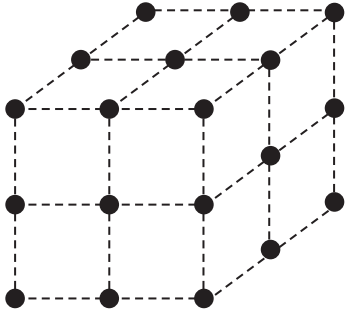
**13** The diagram shows the arrangement of atoms in a particular crystal.



Each atom is at the corner of a cube.

The mass of each atom is $3.5 \times 10^{-25}$ kg. The density of the crystal is $9.2 \times 10^{3}$ kg m$^{-3}$.

What is the shortest distance between the centres of two adjacent atoms?

**A** $3.8 \times 10^{-29}$ m

**B** $6.2 \times 10^{-15}$ m

**C** $3.4 \times 10^{-10}$ m

**D** $3.0 \times 10^{-9}$ m